PHYS 365 - Acoustics, Optics and Modern Physics (for students in Engineering)

COURSE OUTLINE - Fall 2026

A. COURSE INFORMATION

1. Course Description & Learning Outcomes

Please see the university calendar for the course description and pre-requisites:

PHYS 365 https://calendar.ucalgary.ca/courses/1629941

By the end of this course, you will be able to:

- By the end of the course, students will be expected to:
- Exploit and use mathematical representations to predict the behaviour of physical systems exhibiting simple harmonic oscillations or containing waves.
- Analyze and describe the effect of a system's physical properties (density, temperature, mechanical impedance, motion) on wave propagation (speed, reflection, transmission, resonance, and interference.)
- Compute properties and effects of electromagnetic radiation emitted and absorbed by physical systems.
- Extend understanding of classical concepts of particles and waves to the quantum domain.
- Carry out calculations symbolically (in terms of physical variables) and numerically (using appropriate values and their units).
- Obtain experimental data, estimate uncertainties and relate the results to physical laws governing wave behaviour of oscillating systems and waves.
- Collaborate effectively within a team environment and communicate laboratory results in written scientific format.

2. Instructional Team

Role	Name	Get in Touch By
Course Coordinator	Sean Stotyn	phys365369coord@ucalgary.ca
Primary Instructor Lecture 01	Timothy Andrew	timothy.andrew@ucalgary.ca
Primary Instructor Lecture 02	Sean Stotyn	email: sean.stotyn@ucalgary.ca
Lab Coordinator	Timothy Andrew	phys365369coord@ucalgary.ca

3. Course Activities

Activity	Day/Time	Location
Lecture 01	TR 12:30 - 13:45	ENG 60

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Activity	Day/Time	Location
Lecture 02	TR 15:30 - 16:45	ENE 239
Lab 01	T 08:00 - 10:45	ST 036
Lab 02	T 08:00 - 10:45	ST 036
Lab 03	T 15:30 - 18:15	ST 036
Lab 04	T 15:30 - 18:15	ST 036
Lab 05	R 08:00 - 10:45	ST 036
Lab 06	R 08:00 - 10:45	ST 036
Lab 07	R 15:30 - 18:15	ST 036
Lab 08	R 15:30 - 18:15	ST 036
Lab 09	W 11:00 - 13:45	ST 036
Lab 10	W 11:00 - 13:45	ST 036
Lab 11	W 15:00 - 17:45	ST 036
Lab 12	W 15:00 - 17:45	ST 036

Out-of-Class Activity	Day/Time	Location
Midterm Exam	Wednesday, October 21, 2026 at 7:00 pm (90 Minutes)	Details and schedule will be shared on D2L

REGULARLY SCHEDULED CLASSES HAVE PRECEDENCE OVER ANY OUT-OF-CLASS-TIME-ACTIVITY. If you have a conflict with the out-of-class-time-activity, please contact your course coordinator/instructor no later than **14 days prior** to the date of the out-of-class activity so that alternative arrangements may be made.

4. Course Resources

Course Site

PHYS 365 - Fall 2026 -Acoustics, Optics and Modern Physics (for students in Engineering)

Required Textbook

Samuel Ling, Jeff Sanny, & William Moebs, *University Physics, Volumes 1 and 3*: OpenStax.

A set of supplementary course notes will be available on D2L

Mandatory Supplemental Fees

Access to the Achieve platform is required. Access codes can be purchased from the U of C Bookstore, or Achieve can be accessed for free from the TFDL computers.

5. Assessment Components

Assessment	Due Date / Location	Weight
Midterm Exam	Oct 21 2026 Out-of-class	20%
Assignments (x10)	Weekly assignments accessed on Achieve	25%
Laboratory (x5)	A lab schedule will be available on D2L	22.5%
Pre-Lab Quizzes (x5)	5 pre-lab quizzes accessed through D2L	2.5%
Registrar Final Exam	Scheduled by Registrar	30%

Detailed assessment information and expectations can be found on the course D2L.

Group Work Assessment

Lab reports will be graded one per group and each student in the group will receive the same grade for that portion of the lab mark, unless legitimate disputes arise, in which case the lab coordinator will meet with the group to discuss fair grading. Students should attempt to resolve any group conflicts first before contacting the lab coordinator.

6. Letter Grade Conversion

Letter Grade	A+	A	A-	B+	B	B-	C+	C	C-	D+	D	F
Minimum Percent Required	95	90	85	80	75	70	65	60	55	50	45	< 45%

7. Missed Course Assessments

We understand that unexpected circumstances can arise during the term. The table below outlines what to do in different situations so we can help you as effectively as possible.

Assessment	What to Do	Options/Next Steps

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Assignments (x10) , Laboratory (x5) , Pre-Lab Quizzes (x5)	**Missed Assignments**: There will be no opportunities to make up for missed assignments, and the assignment deadlines will not be extended, barring extenuating circumstances that are approved by the course coordinator on a case-by-case basis. **Missed Lab**: Students who miss a lab should fill in the Missed Lab Form posted on D2L and submit it to the appropriate dropbox within **5 business days** of missing the lab. **Missed Pre-Lab Quiz**: The lowest pre-lab quiz grade will be dropped. If more than one pre-lab quiz is missed, the student should contact the course coordinator.	**Assignments**: Each assignment will open on a Wednesday at 5:00 PM and close the Wednesday two weeks later at 11:59 PM. Students are advised to complete each assignment within the first week they are released. **Labs**: The lab coordinator will get back to the student regarding an alternative arrangement, which may include shifting the weight to the other labs, shifting the weight to the final exam, or arranging for a deferral of term work. A missed lab that does not have an approved accommodation from the course coordinator will result in a grade of zero. **Pre-lab**: If an alternative arrangement is approved by the course coordinator, the weight of the pre-lab quiz will be shifted to the corresponding lab.
Midterm Exam	If a student misses the midterm exam, they should fill out a Missed Midterm Form and submit it to the appropriate dropbox within **5 business days**.	There is no deferred midterm exam in this course; If the request is approved, the weight of the midterm will be shifted to the final exam. A missed midterm that does not have an approval from the course coordinator will result in a grade of zero.
Final Exam	If you miss a Registrar-scheduled final exam, you must follow the Deferred Final Exam (DFE) process. Begin by visiting the Registrar's website for full instructions.	The Registrar will communicate with you regarding the status of your deferred exam application. If approved, the Registrar will post the date, time, and location of your deferred exam in your Student Centre.

Extended Absence / Significant Disruption: If you experience an interruption that affects multiple assessments or several weeks of learning, additional support may be needed. Please inform the instructional team via email as soon as you are able. You can consult the UCalgary Course Interruption resource for guidance: https://www.ucalgary.ca/current-students/course-interruption

B. COURSE POLICIES

1. Academic Integrity

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Academic integrity is a core part of your learning and of our shared academic community. All work you submit must represent your own thinking and effort, unless collaboration or specific tools are explicitly allowed. Academic misconduct includes (but is not limited to) plagiarism, cheating, unauthorized collaboration, and the use of unapproved aids.

Below are the academic integrity expectations specific to this course:

Acceptable & Prohibited Tools & Resources

Generative AI and LLMs are tools that are increasingly being integrated into professional life. This course allows the use of AI to help students prepare lab reports, in whole or in part. It is the student's responsibility to ensure that the information provided by the AI tool used is accurate, at an appropriate level for the course, consistent with the methods and notation being used in the course, and free of any AI generation errors, such as non-existent references. If AI tools are used, it should be declared on the cover page of the lab report, stating which tools and how they were used.

When completing homework problems, the homework platform Achieve has a built-in AI chatbot tutor that students are encouraged to use when they get stuck. Students should not use additional AI tools to complete their homework.

Exam Policy

All exams in this course are closed book, in person, multiple choice tests. A formula sheet will be provided in the exam booklet and it will be posted to D2L at the beginning of the semester so that students can familiarize themselves with it. Only a Schulich approved calculator is permitted during examinations; programmable and graphing calculators are not permitted, nor are devices capable of communicating, such as Smart watches and AI-integrated glasses.

Resources & Support

The Student Success Centre offers workshops, tools, and a student academic integrity handbook to help you understand expectations and avoid common pitfalls. They also provide guidance on citation, collaboration, and working with sources. Learn more at: ucalgary.ca/student-services/student-success/learning/academic-integrity

If you ever have questions about how to apply academic integrity principles in this course, please reach out. We are here to help you navigate expectations and develop strong academic skills.

2. Grade Reappraisal

If you believe there has been an error in the marking of your work, you may request a grade reappraisal. The goal of this process is to ensure fairness and accuracy while maintaining consistent academic standards.

Formal Grade Reappraisal Request

If concerns remain after discussions with the course instructor, you may submit a formal reappraisal request. A formal reappraisal may result in the grade being raised, lowered, or left unchanged. Please see the Faculty of Science guidelines on the grade reappraisal process: science.ucalgary.ca/current-students/undergraduate/science-advising/grade-reappraisals-and-appeals

C. SUPPORTS

You don't have to navigate university alone. Here are resources available to support your success and well-being:

Undergraduate Science Centre(USC)	Support for program planning, course selection, and academic regulations
Student Wellness Services	Physical and mental health support, including counselling, medical services, and wellness programs. ucalgary.ca/wellness-services/services
Student Success Centre	Help with study skills, writing support, time management, and academic strategy development. ucalgary.ca/student-services/student-success
Libraries and Cultural Resources	Research support, access to scholarly resources, citation help, and subject?specific librarians. library.ucalgary.ca
Safety Resources	Safewalk: 403-220-5333 I Campus Security: 403-220-5333 Available 24/7 for safety escorts, emergency response, and campus support.

For a comprehensive list of student wellness and safety resources, please visit: ucalgary.ca/registrar/registration/course-outlines

D. UNIVERSITY POLICIES

For detailed university policies, please consult the Academic Calendar at calendar.ucalgary.ca

1. Academic Accommodations

We are committed to supporting your learning. If you require academic accommodations, please follow the University of Calgary's established policies and procedures. Reaching out early helps ensure that the appropriate supports can be arranged.

Accommodations for Disability or Medical Conditions

Students who require accommodations related to a disability or ongoing/temporary medical condition should contact Student Accessibility Services (SAS). SAS works with students and instructors to set up academic accommodations such as extra time, note-taking support, or exam adjustments. Website: ucalgary.ca/student-services/access/

Accommodations Based on Other Protected Grounds

Students seeking accommodations based on protected grounds other than disability (e.g., religion, family status, gender expression or identity) should submit a Request for Accommodation in Academic Courses Form. Please email the completed form to science@ucalgary.ca at least 10 business days before the relevant assessment or planned absence.

2. Copyright and Intellectual Property

Course materials (including lectures, notes, assignments, exams) are the intellectual property of the instructor and are protected by copyright. Unauthorized sharing, distribution, or posting of course materials is prohibited and may result in disciplinary action under the Non-Academic Misconduct Policy. Students must read and comply with the University's Acceptable Use of Material Protected by Copyright Policy: ucalgary.ca/legal-services/university-policies-procedures/acceptable-use-material-protected-copyright-policy

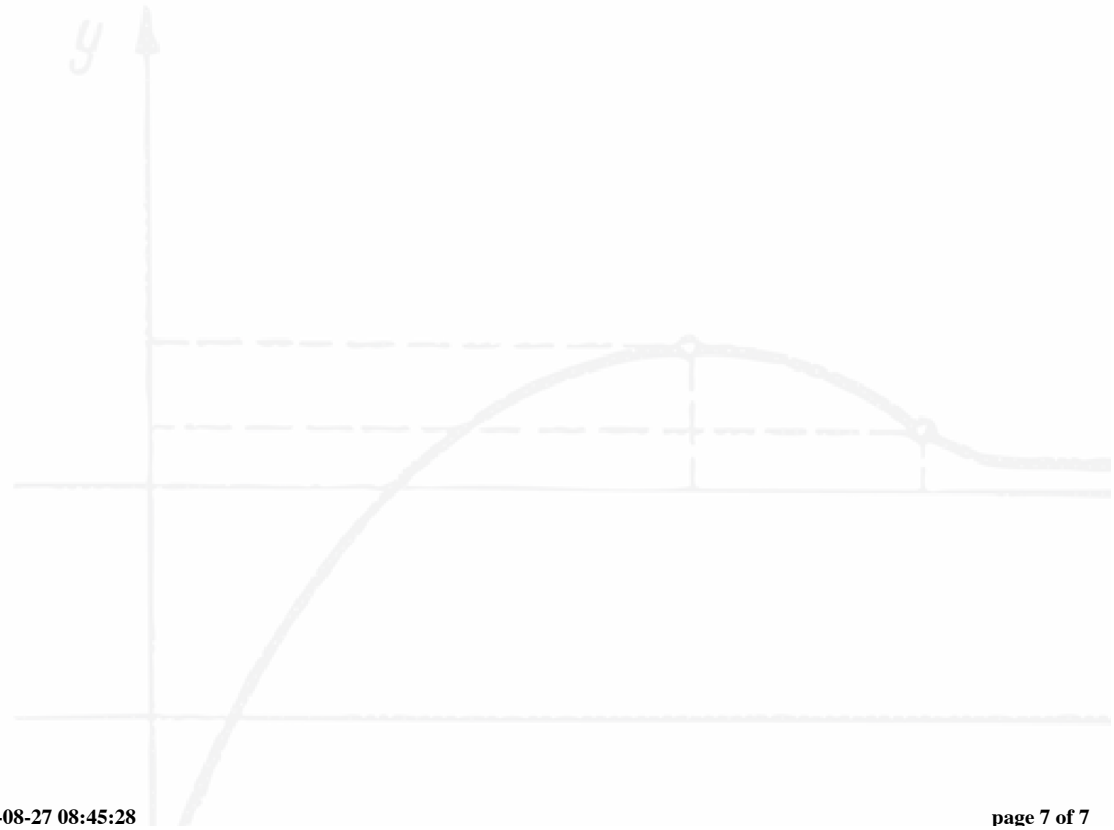
3. UCalgary Notice of Collection

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The University of Calgary (UCalgary) respects your privacy and is committed to ensuring the privacy of all students, staff, and community members. Your personal information is being collected under the authority of section 4(c) of the Protection of Privacy Act (POPA). It will be collected, used and disclosed for the purposes outlined in UCalgary's Notice of Collection, Use and Disclosure of Personal Information and managed in accordance with UCalgary's Privacy Management Program. Your personal information may be input into automated systems to generate content or make decisions, recommendations, or predictions. Assignments can only be returned to the student and will be accessible only to authorized faculty and staff. If you have questions regarding this notice or UCalgary's collection, use, or disclosure of your personal information, please contact accessandprivacy@ucalgary.ca.

Approval Date: Aug 26 2026 22:08

The University of Calgary, located in the heart of Southern Alberta, both acknowledges and pays tribute to the traditional territories of the peoples of Treaty 7, which include the Blackfoot Confederacy (comprised of the Siksika, the Piikani, and the Kainai First Nations), the Tsuut'ina First Nation, and the Stoney Nakoda (including Chiniki, Bearspaw, and Goodstoney First Nations). The City of Calgary is also home to Métis Nation within Alberta (Nose Hill Métis District 5 and Elbow Métis District 6).



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